

```
C:\WINDOWS\system32\cmd. x + v

[notice] To update, run: python.exe -m pip install --upgrade pip

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>python -c "import pandas, sklearn, mlflow, fastapi, pytest, joblib; print('All packages OK!)"
All packages OK

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>notepad app\config.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>notepad app\data_loader.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>notepad app\preprocessing.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>notepad app\preprocessing.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>python -c "from app.data_loader import load_clickstream, load_product_reviews, load_ecommerce_tests; print('Clickstream:', load_clickstream().shape); print('Reviews:', load_product_reviews().shape); print('Test cases:', len(load_ecommerce_tests()))"
Clickstream: (1000, 16)
Reviews: (800, 11)
Test cases: 50

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>python -c "from app.data_loader import load_clickstream; from app.preprocessing import clean_clickstream, prepare_features; df=clean_clickstream(load_clickstream()); X,y,p=prepare_features(df); print('Cleaned:', df.shape); print('Features:', X.shape); print('Target:', y.shape); print('Preprocessing pipeline: OK!)"
Cleaned: (1000, 16)
Features: (1000, 9)
Target: (1000,)
Preprocessing pipeline: OK

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>notepad app\baseline_model.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>python app\baseline_model.py
Traceback (most recent call last):
  File "C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02\app\baseline_model.py", line 19, in <module>
    from app.data_loader import load_clickstream
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(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>notepad app\baseline_model.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>python -m app.baseline_model
Loading clickstream dataset...
Original dataset shape: (1000, 16)
Cleaned dataset shape: (1000, 16)
Feature records: (1000, 9)
Target records: (1000,)
Training records: 800
Testing records: 200

Starting baseline model training...

=====
OmniRetail AI Baseline Model
=====

Dataset records : 1000
Training records: 800
Testing records : 200
Feature count : 9

Evaluation Metrics
Accuracy : 0.4900
Precision: 0.5306
Recall : 0.7027
F1 Score : 0.6047
ROC-AUC : 0.4498

Classification Report
precision recall f1-score support
0 0.38 0.22 0.28 89
1 0.53 0.70 0.60 111
accuracy 0.49 200
```

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C:\WINDOWS\system32\cmd. x + v

    accuracy          0.49      200
    macro avg         0.45      200
    weighted avg      0.46      200

Model saved : C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02\models\baseline_purchase_model.j
oblib
Metrics saved : C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02\docs\baseline_metrics.json
MLflow DB : C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02\mlflow.db

Baseline model completed successfully.
=====

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>notepad tests\test_baseline_model.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>python -m pytest tests -v
===== test session starts =====
platform win32 -- Python 3.12.5, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI
-Checkpoint-02\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02
plugins: anyio-4.15.1
collected 7 items

tests/test_baseline_model.py::test_clickstream_dataset PASSED [ 14%]
tests/test_baseline_model.py::test_product_reviews_dataset PASSED [ 28%]
tests/test_baseline_model.py::test_ecommerce_test_cases PASSED [ 42%]
tests/test_baseline_model.py::test_preprocessing_pipeline PASSED [ 57%]
tests/test_baseline_model.py::test_saved_baseline_model PASSED [ 71%]
tests/test_baseline_model.py::test_baseline_metrics_file PASSED [ 85%]
tests/test_baseline_model.py::test_mlflow_database PASSED [100%]

===== 7 passed in 5.41s =====

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>notepad docs\BASELINE_REPORT.md

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>dir docs
```

```
C:\WINDOWS\system32\cmd. x + v

Directory of C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02\docs
05-10-2026 18:55 <DIR> .
05-10-2026 18:54 <DIR> ..
05-10-2026 18:53 357 baseline_metrics.json
05-10-2026 18:55 7,316 BASELINE_REPORT.md
                2 File(s) 7,673 bytes
                2 Dir(s) 9,072,738,304 bytes free

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>notepad README.md

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>notepad requirements.txt

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>dir
Volume in drive C is Windows-SSD
Volume Serial Number is EA8D-9B22

Directory of C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02
05-10-2026 18:57 <DIR> .
05-10-2026 08:52 <DIR> ..
05-10-2026 18:54 <DIR> .pytest_cache
05-10-2026 18:36 <DIR> .venv
05-10-2026 18:48 <DIR> app
05-10-2026 08:54 <DIR> data
05-10-2026 18:55 <DIR> docs
05-10-2026 18:53 876,544 mlflow.db
05-10-2026 18:53 <DIR> mlruns
05-10-2026 18:51 <DIR> models
05-10-2026 08:53 <DIR> notebooks
05-10-2026 18:56 6,313 README.md
05-10-2026 18:57 92 requirements.txt
05-10-2026 18:54 <DIR> tests
                3 File(s) 882,949 bytes
               11 Dir(s) 9,061,228,544 bytes free
```

```
C:\WINDOWS\system32\cmd. X + v

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>python -m pytest tests -v
===== test session starts =====
platform win32 -- Python 3.12.5, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02
plugins: anyio-4.15.1
collected 7 items

tests/test_baseline_model.py::test_clickstream_dataset PASSED [ 14%]
tests/test_baseline_model.py::test_product_reviews_dataset PASSED [ 28%]
tests/test_baseline_model.py::test_ecommerce_test_cases PASSED [ 42%]
tests/test_baseline_model.py::test_preprocessing_pipeline PASSED [ 57%]
tests/test_baseline_model.py::test_saved_baseline_model PASSED [ 71%]
tests/test_baseline_model.py::test_baseline_metrics_file PASSED [ 85%]
tests/test_baseline_model.py::test_mlflow_database PASSED [100%]

===== 7 passed in 4.68s =====

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>dir models
Volume in drive C is Windows-SSD
Volume Serial Number is EA8D-9B22

Directory of C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02\models

05-10-2026 18:51 <DIR> .
05-10-2026 18:58 <DIR> ..
05-10-2026 18:53 5,322 baseline_purchase_model.joblib
1 File(s) 5,322 bytes
2 Dir(s) 9,058,410,496 bytes free

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>dir docs
Volume in drive C is Windows-SSD
Volume Serial Number is EA8D-9B22

Directory of C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02\docs
```

```
C:\WINDOWS\system32\cmd. X + v

2026/10/05 19:01:07 INFO: Finished server process [33900]
2026/10/05 19:01:07 INFO: Received SIGINT, exiting.
2026/10/05 19:01:07 INFO: Terminated child process [33900]
2026/10/05 19:01:07 INFO: Terminated child process [29624]
2026/10/05 19:01:07 INFO: Terminated child process [27628]
2026/10/05 19:01:07 INFO: Terminated child process [23480]
2026/10/05 19:01:07 INFO: Waiting for child process [33900]
2026/10/05 19:01:07 INFO: Waiting for child process [29624]
^C
(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>type docs\baseline_metrics.jsontype docs\baseline_metrics.json
The system cannot find the file specified.
Error occurred while processing: docs\baseline_metrics.jsontype.

docs\baseline_metrics.json

{
  "model": "Logistic Regression",
  "dataset": "aiml_ecommerce_clickstream.csv",
  "total_records": 1000,
  "training_records": 800,
  "testing_records": 200,
  "feature_count": 9,
  "target": "label_purchased",
  "accuracy": 0.49,
  "precision": 0.5306,
  "recall": 0.7027,
  "f1_score": 0.6047,
  "roc_auc": 0.4498
}

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02\.venv\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_envirom
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02\.venv\Lib\site-packages\mlflow\server\fastapi_ap
```

```
C:\WINDOWS\system32\cmd. x + v
2026/10/05 19:01:13 INFO: Application startup complete.
2026/10/05 19:01:13 INFO: Shutting down
2026/10/05 19:01:13 INFO: Waiting for application shutdown.
2026/10/05 19:01:13 INFO: Application shutdown complete.
2026/10/05 19:01:13 INFO: Finished server process [29624]
2026/10/05 19:01:13 INFO: Waiting for child process [27628]
2026/10/05 19:01:13 INFO: Waiting for application shutdown.
2026/10/05 19:01:13 INFO: Application shutdown complete.
2026/10/05 19:01:13 INFO: Finished server process [27628]
2026/10/05 19:01:13 INFO: Waiting for child process [23480]
2026/10/05 19:01:13 INFO: Stopping parent process [32908]

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>type docs\baseline_metrics.json
{
  "model": "Logistic Regression",
  "dataset": "aiml_ecommerce_clickstream.csv",
  "total_records": 1000,
  "training_records": 800,
  "testing_records": 200,
  "feature_count": 9,
  "target": "label_purchased",
  "accuracy": 0.49,
  "precision": 0.5306,
  "recall": 0.7027,
  "f1_score": 0.6047,
  "roc_auc": 0.4498
}
(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>python -c "import joblib; from app.data_loader import load_clickstream; from app.preprocessing import clean_clickstream, prepare_features; model=joblib.load('models/baseline_purchase_model.joblib'); df=clean_clickstream(load_clickstream()); X,y,_=prepare_features(df); print('Loaded model:', type(model).__name__); print('Sample predictions:', model.predict(X.head(10)).tolist()); print('Prediction test: PASSED')"
Loaded model: Pipeline
Sample predictions: [1, 1, 1, 1, 0, 1, 1, 1, 1, 1]
Prediction test: PASSED

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\OmniRetail_AI-Checkpoint-02>
```